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SOFTWARE ENGINEER II | Reports to: Shashank Rachamalla

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Reflection Period: Nov 15, 2025 - May 15, 2026

Impact is **what** you accomplish - and **how** you do it matters just as much.

WHAT

What results have you delivered for your goals, keeping security, quality, and AI in mind?



HOW

How have your behaviors and actions led you, your team and Microsoft to excel, grow, and build trust?

Reflect on the past

What results did you deliver, and how did you do it?

#	Feature/Area	What was delivered & how	Impact/Results
Feature			
1	Velero + AKS Backup Extension Upgrade	• Delivered Velero and AKS Backup Extension upgrade as a **one-time major engineering effort**. • **Reorganized BCDR custom** changes and standardized upgrade pipelines for **easier future upgrades**. • Fixed **memory leaks, CSI leaks, and performance issues.** • Broad refactoring for maintainability. • Rolled out across all regions with **0 regressions**.	• Improved Velero backup and GC performance. • Reduced OOM kills via memory-leak fixes. • Enabled Parallel Item Block backup support. • Improved logging fidelity and per-object restore status visibility. • Added Azure certificate-based auth. • Improved CSI snapshot behavior. • **Resolved 28 CVEs**, bringing extension vulnerabilities to 0. ***(Growth Mindset, Accountability)***
2	Direct Drive Support for AKS and Disk (Completed)	• Completed **Direct Drive implementation** across AKS and Disk workloads. • Delivered architecture across management, control, and data plane changes. • Executed policy and flow readiness for production rollout.	• Improved Disk backup **RPO** from **12h to 4h**. • **80+** subscriptions onboarded. • **800+** Direct Drive disks backed up. • Enabled **new customer adoption** where Direct Drive was the primary blocker. ***(Customer Obsessed, Accountability)***
3	Modify Protection + AFS PuPr (Public Preview) with CLI and Portal Support	• Delivered **Modify Protection** with AFS PuPr end-to-end across backend, Portal, and CLI. • Rolled out with **0 regressions.** • Portal fixes: dirty state tracking, secrets checkbox priority, empty kind validation. • Cross-workload handler fixes (applies to all workloads).	• **2K+** Modify BI calls performed. • **100+** customers using Modify Protection. • **~20%** brownfield customers enabled AFS on existing workloads. • Portal/CLI fixes reduce friction across all workloads. ***(Customer Obsessed, Accountability)***
4	AAD DataSource Support During Hardening (In Progress)	• Closed design discussions with multiple partner teams and finalized scope. • Created PRs currently in review for rollout. • AAD token handling fixes.	• Positions platform for stronger authentication reliability. • Enables easier extensibility in upcoming releases.

		• PageBlob client updated to use auth mode. • Extensibility improvements for future auth scenarios.	
Quality+Productivity			
1	Production Release Ownership and SLA	• Drove **4 major releases**: Scenario Plugin Host, AKS Backup Extension, and 2 Retention Agent releases (2601/2604/2604-2/2605). • **120+** completed ADO PRs across core repos. • Maintained FI/RI readiness, rollback clarity, changelog hygiene, Safe-Fly/CRB discipline. • Fixes: CreateConfigMapForRestore error handling. • TA Errors mapping for AKS operations. • RAS DM Version upgrade to fix restore-success false positives. • CASC handling updates for lower RPO. • AAD-enabled snapshot support during hardening. • PV2 vault restore region validation. • Missing Activity Stats telemetry fix.	• Restored AKS workload SLA to **99.9%.** • Positioned RA hardening **toward 99.9%.** • Improved customer actionability via better user error codes. • **Reduced operational noise** and false-positive service errors. ***(Accountability, Customer Obsessed)***
2	BVT and Validation Framework Expansion	• **Expanded BVT coverage** for Direct Drive and Disk scenarios. • **Enabled soft-deletion support** in BVT and runners. • Created runbooks for resource **cleanups** and **CVE patch** deployments. • Added critical validation to **prevent false positives**. • **Fixed broken runners** and enabled monitors for AKS Configure and CRR. • Improved cleanup stability and test filtering controls. • Enabled runner readiness for broader regional coverage.	• Improved test reliability and reduced gate flakiness. • **Strengthened validation** signal confidence. • Reduced operational overhead for long-running resources. • Enhanced runner monitoring ensures accurate regression detection. ***(Growth Mindset)***
3	Upstream OSS Contributions and Design Leadership	• **18** authored PRs upstream in velero-io/velero. • **VSC cleanups** and CSI lifecycle fixes. • Backup **deletion correctness** improvements. • Multi Hooks error handling. • Multi-branch backports for alignment. • Design document for **restore-created VSC cleanup** (in review).	• **Strengthened Azure and upstream** Velero reliability paths. • VSC cleanup design prevents operational issues at scale. • Demonstrated sustained external technical leadership. • **Increased Azure influence** in the Velero community. ***(One Microsoft, Growth Mindset)***
Supportability			
1	Error-Code Hardening and Diagnosability	• Implemented actionable error-code mappings and fail-fast behavior. • Private endpoint tiering failures. • SHIM in-tree volume handling. • CSI authorization short-circuiting. • HNS unsupported account errors. • Target-cluster stopped-state errors. • New PM-approved user error codes for multiple failure scenarios. • K8s configure backup validations. • PV2 vault restore region checks. • Managed Disk operation error mappings.	• **Reduced triage ambiguity** for support and field teams. • Improved customer **self-service troubleshooting**. • **False service-error** reports **reduced**. • **SLA measurement accuracy** improved. *(Customer Obsessed)*
2	Operability and CSS assistance/ CRI Resolutions	• Conducted AKS hardening bug reviews with Prot and DM teams. • Fixed false positives in RAS. • Resolved VSC missing/stalled scenarios (ArgoCD conflicts, GC errors). • Identified and remediated validation gaps in Backup Agent and BMS. • Resolved 1 refund-level CRI with code patch. • Clarified supported vs. unsupported scenarios to prevent escalations.	• **Reduced customer escalations** and improved operational clarity. • **CRI resolution** prevented customer friction. • Enabled team knowledge transfer for advanced troubleshooting.
Security			
1	Vulnerability Remediation and Hardening	• Resolved **28 CVEs** in Velero and plugin paths. • Brought Velero image vulnerabilities to **0**. • Coordinated **S360 warning resolution.** • Implemented credential leak fixes via credscans and log sanitization.	• Eliminated security exposure across backup extension supply chain. • Strengthened secure-release posture and customer trust. ***(Accountability)***
2	MSRC Risk Response and Endpoint Validation	• Responded to **storage endpoint token-exfiltration risk**. • Implemented validation and allowlisting controls to prevent unsafe redirects.	• Closed credential leakage risk vector. • Improved customer security posture in extension workflows.

Leveraging Others / AI			
1	Technical Alignment and AI-Driven Automation	• Aligned AKS, Disk, CSI, and release stakeholders through design walkthroughs and shared artifacts. • Enhanced BCDR AI Agent skills for AKS and MSR workloads. • Applied MCP Servers for design workflows and diagrams. • Leveraged RTI MCP Agents for automated log analysis. • Used PR Assistant for routine fixes	• Reduced cross-team rework and accelerated decision velocity. • AI transformed **2–3-day** manual investigations into **hours.** • Significantly improved team productivity through automation. ***(One Microsoft, Growth Mindset)***
Helping Others			
1	Release and Hotfix Shepherding, Code Review Mentoring, and CRB Collaboration	• **57 engaged reviews** concentrated on peers and developing engineers. • Prioritized functional correctness, reliability, and release safety. • Provided scripts and testing **assistance in CRB** for UAE Customers. • Resolved SDK bug in Azure ResourceManager blocking cross-team work.	• Increased team quality bar and accelerated teammate delivery. • Reduced context gaps and onboarding friction. • Improved operational confidence and reduced release stress. • SDK fix unblocked downstream teams on dependencies. ***(One Microsoft, Accountability)***

Reflect on recent setbacks - what did you learn and how did you grow?

During the **SPH 2601 release**, an idle code path in the restore flow was unexpectedly triggered, causing **restore failures**. I caught it through proactive monitoring and quickly executed a **multi-region hotfix** without escalation. This taught me two critical lessons: **comprehensive corner-case test coverage** is essential because gaps leave dormant code paths unvalidated until production, and that **failures in complex systems are inevitable—**what matters is staying calm, triaging systematically, and executing fixes with confidence rather than reacting in panic.

A second challenge emerged during the Velero 1.16 upgrade: **building consensus on complex decisions** in upstream under time pressure. I discovered that technical correctness alone is insufficient; stakeholders need the *why* communicated with context, not just the *what.*
The lesson I carry forward is that influence across team boundaries requires as much **attention to communication design as to engineering design.**

Manager comments

Priyansh has delivered **strong, high-impact contributions across AKS Backup, Vault-tier, and Ops-tier**, demonstrating ownership, technical depth, and ability to execute complex work end-to-end at scale.
At an individual level, he has **driven multiple critical features that directly unlocked customer adoption and platform scale**. He led the design and delivery of **AKS Direct Drive backup/restore (PV2 & Ultra disks)**, enabling **80+ subscriptions and 800+ disks**, opening up scenarios where backup was previously blocked. He also delivered **AKS Modify Protection and AFS Ops-tier Public Preview**, driving **100+ active customers and ~20% brownfield adoption**. In addition, he proactively drove **all-region enablement across 11 regions** by upgrading AKS ARM SDK dependencies across multiple repos, removing a key onboarding and scaling bottleneck.
Priyansh has also demonstrated strong impact in **driving reliability and customer experience improvements**. He resolved deep system issues such as trusted access resiliency, eliminating **2–3 recurring customer/BVT failures per month**, and improved **backup success and clarity through better error mappings**, leading to a **0.2% SLA improvement**. He addressed critical performance issues, including **reducing GC time by 50%**, improving **RPO from 12 hours to 4 hours for PV2/Ultra disks**, fixing memory consumption in large clusters, and ensuring reliable multi-restore scenarios. These changes significantly improved system stability and customer trust.
On fundamentals, Priyansh has consistently **raised the bar on quality, test reliability, and engineering productivity**. He strengthened **Vault-tier BVT coverage (IR/DR/CRR)**, reduced false positives, and ensured stable validation signals for releases. He improved **merge validation time by ~30 minutes**, accelerating developer iteration cycles, and built automation for **resource cleanup, test cluster maintenance, and operational workflows**, reducing manual overhead and improving team efficiency.
He has also demonstrated an **upstream-first mindset and ecosystem impact**, successfully driving the **Velero upgrade (1.13 → 1.16)** with upstream fixes, improving maintainability and reducing long-term technical debt. His work on building a **Velero E2E validation pipeline** helped identify and fix Azure-specific issues early, benefiting both internal quality and the broader community.
Priyansh consistently shows **strong ownership, cross-team collaboration, and initiative**, driving work across AKS, Disk, CSI, and OSS teams with minimal supervision. He proactively takes on critical gaps, ensures end-to-end closure of complex problems, and contributes to overall team success through knowledge sharing and support during bandwidth constraints.

Details (Keep it crisp and tuned to presenting data versus adjectives. It should be self-explanatory for anyone outside the area and not required to be presented by the manager)

PR data (excluding velero repo) - 6 months

Alias	ManagerEmailName	CountPRsReviewed_Last6Months	CountComments_Last6Months	PRsMeetThreshold_Last6Months	CommentsMeetThreshold_Last6Months	CountPRsReviewed_Las
choudharypr	shrac	55	31	10	29	107

AR data (May'25 to May'26)

Circle of Impact	Short description of work	Impact/Results and HOW (OKR, SLA improvement, Customer Obsession, Growth Mindset etc.)	Impact Rating
IC - Feature	• **AKS Direct drive support** - Designed and implemented direct-drive (PV2 and Ultra disk) backup and restore support across AKS and Disk. • **Velero E2E pipeline** - Designed & Implemented an E2E test pipeline dedicated to validating Azure-specific functionality in velero upstream. Run daily with results published to upstream. • **AKS Modify Protection** - Designed and implemented AKS Modify Protection with end-to-end CLI and Portal support. • **AKS AFS PV Ops-tier** - Drove AKS AFS PV Ops tier Public Preview (PuPr) delivery with CLI and Portal integration. • **Enabled all-region support** for AKS ops-tier and vault-tier by upgrading AKS ARM SDK across Common, SPH, Store, and Test repos to support (as new regions only supported newer SDK)	**AKS Direct drive** • **80**+ subscriptions onboarded. • **800**+ Direct Drive disks backed up. • Enabled **new customer adoption** where Direct Drive was the primary blocker. ***(Customer Obsessed, Accountability)*** **Zero regressions** **Velero E2E pipeline** We've identified and fixed **multiple Azure-specific bugs** in Velero, improving stability for all users. Key ask from community to avoid Azure regressions. **AKS Modify protection** **2K**+ Modify BI calls performed. • **100**+ customers using Modify Protection. **AKS AFS Ops tier PuPr** **~20%** brownfield customers enabled AFS on existing workloads. **All-region support** Key onboarding blocker. Opened up 11 regions where AKS had support but backup not present.	Exceptional Volume of work. Strong sense of accountability. Proactively picked up All region support as a stretch Handled partner collab exceptionally well at his level across multiple teams and OSS teams with little involvement from other team members. Drove portal changes for modify protection and AFS public preview.
IC - Fundamentals Quality/Productivity	**Velero Upgrade in AKS Backup Extension** • Upgraded Velero from 1.13 → 1.16 in AKS Backup Extension; multiple upstream fixes/optimizations delivered with zero regressions • Enabled legacy VSC cleanup for backward compatibility; removed orphaned snapshots and reduced customer cost (upstream patch) • Improved GC speed by 50% for large clusters by removing unnecessary Worked with CSI diver team in handling the CSI cache issue which need adding a small delay between GC calls for driver < 1.33 • Added retries in the GC flow in fetching snapshot info tar file form snapshot storage account (CRI with refund) • Added fail-fast for permission/user errors in async flows added for PV2/Ultra support • Added MV/unit tests for BCDR forks (Velero + Azure plugin / object store) • Simplified future upgrades: consolidated all BCDR changes into a single commit on top of 1.16 (much easier upgrade now) • Fixed upstream tracking issue (1.15/1.16): reduced stale resource tracking via hook timeout (10 min), pre-checks, and faster job termination **Quality** • **Improving Disk Opstier RPO for PV2/Ultra** - we now support 4hrs from earlier 12hrs with changes in Timeout and BA concurrency settings and clear user messaging. • **Error mappings -** Sideloaded velero, private disks, In-tree persistent volumes, CSI snapshot user errors, HNS storage accounts not supported, etc. • **Argo CD whitelisting -** Argo CD was cleaning up the volume snapshots in cluster (which we were creating as part of backup). Worked with velero upstream to put a specific velero label and shared guidance to customers to exclude this in their Argo CD config. • **Restoring twice to same cluster was failing** due to older validation objects lying on the cluster. Fixed to clean up the older objects and keep cluster state fresh for subsequent restore.	• GC speed improved by 50% • GC much more resilient to transient errors in customer's cluster • Better MV coverage for OSS forks repo and easier upstream upgrades with consolidation of local commits • Better handling of orphaned snapshots in upgrade scenarios to avoid unnecessary retention cost to customers. • Faster response time to customers for backup failures pertaining to user errors. • Contributions were acknowledged resulting in **promotion to the Velero Triage role**. • Easier future upgrades with consolidation of local commits • Better RPO to PV2/Ultra disk customers. • Reduction in CRIs for Sideloaded velero, ARGO CD and In tree warning scenarios. • No more restore failures when customer tries to restore multiple times to same target • Zero backup failures due to TA throttling/ambassador issues	Exceptional Volume & Agility Upstream mindset than patching locally Strong accountability Proactively picked up multiple items and often stretched to meet the commitment around security vulnerabilities to end users. Leveraged PRISM AI TestGen, given full flow context, generated a test matrix covering corner cases like partial GC failures, concurrent VSC deletion races, and hooks-discarded scenarios. • Caught 2-3 bugs in GC flows pre-release that BVTs wouldn't have surfaced until production. Leveraged AI Customer tool executed scenarios in parallel across configurations, producing results quickly.

	• Strengthened **Trusted Access calls resilience -** Transient failures at the TA end points. We added exponential retries and also better mapping of headers to error codes. • Fixed List CRDs memory-consumption issue in the extension for large clusters. Moved to GET calls instead. • Cut SPH merge-validation time by 30 min for faster feedback. (was 1hr before this) **BVT Improvements** • Added IR, DR & CRR tests for Vault tier BVTs - Stabilized them - overall they are running smooth. • Support for vault soft delete in BVTs. Using Guids for BIs , cleaning up BIs instead of vaults, etc. • Proactively Fixing false positives in BVTs. Some DPP workloads have succeeded with warnings. BVTs moved to work on full success scenarios. • Fixed cluster level post validations in AKS where we were triggering async without await. Created runbooks for resource **cleanups** and **CVE patch** deployments to automate k8s version on long haul cluster setups **Security** • MSRC case - Implemented validation and allowlisting controls to prevent unsafe redirects. Eliminates token exfiltration where user can provide invalid storage account URLs.	• Merge validation time improved by 30 mins for SPH builds. • 2 - 3 BVT/customer backup failures per month to zero failures after improving resiliency for trusted access gateway calls. • Improved SLA by 0.2% through better user error mappings • Fixing key false positives in BVTs for true cover. We had scenarios in past where we had CRIs due to succeeded with warning cases. • Automated cleanup and patching for test clusters - Dev cycles saved	
Helping others	**Helping Team and Others** • Helped Anshul with private stamp B and C setups for SPH to increase team productivity. • Created scripts for bug bash setups and one-click deploys of end-to-end AKS flows for testing. • Helped Hariharan with debugging PV2 Resize issues. • Helped Ankur with runners and BVT flow updates for BMS features. • Helped Anshul in fixing runners for AKS and Disk flows. • Fixed quota issues and telemetry fixes for BVTs and Runners. • Provided scripts and testing assistance in CRB for UAE customers. • Resolved SDK bug in Azure ResourceManager NuGets that was blocking cross-team deliverables. **Dashboards, TSGs & Knowledge Transfer** • Improved Vault stats dashboards for team visibility. • Authored and refined TSGs for ReIR/CRI/debugging workflows. • Created TSGs for CSS folks on new unique CRIs to reduce repetitive debugging and escalation cycles. • Onboarded AKS Vault Tier to MSR and created dashboards to help the team track health. • Helped team understand velero and AKS backup flow in weekly learning sessions. **AI & Automation Enablement** • Enhanced BCDR AI Agent skills for AKS and MSR workloads — enabling teammates to use AI-driven analysis for routine investigations. **Debugging, CRI & On-Call** • Helped Mayank and Shashank with debugging long-running CRIs for AKS. • Took up multiple AvA asks to support the team during high-pressure periods. • Took SME on-call duty for three continuous weeks to support team. • Discussed security issues with Yasin around Shim and Velero OSS Pipeline.		Exceptional. - Stepped up multiple times to help folks.
Leveraging Others	• Used Anshul's one-prompt deployment pipeline for private stamp deployments and testing — reduced manual setup and accelerated validation cycles. • Used skills and agents created by Mayank for operability analysis and monthly MSR analysis — turned multi-hour debugging into structured, repeatable workflows.		

	• Built design documents on top of existing AKS and Disk flow docs — leveraged prior architecture artifacts to accelerate design decisions and reduce duplication of effort. • Leveraged Harness engineering demonstrated by Mayank in weekly learning sessions — applied patterns to improve workflow efficiency. • Leveraged AI customer tool built by Arvind to test flows during bug bashes — accelerated validation without manual setup. • Used PRISM AI TestGen to build test plans for feature work — reduced test planning effort and improved coverage.		

Areas of improvement: He has a great attitude, and I have never seen him say no to any work that comes to his plate. This is a good trait but sometimes I feel he picks up too many things on this plate, and this might burn him off. I have asked him to watch out or monthly commitments before picking more work and load balance accordingly without burnout.

Plan for the future

What are your goals for the upcoming period?

Goal #1. AFS Vault Tier Support

- Enable backup and restore of AFS volumes in AKS to the Vault tier with smooth customer experience across different specifications and workload types.
- Target zero regressions to existing AKS workflows and ensure comprehensive test coverage for these new scenarios before GA.
- Build the foundation for extending Vault tier support to additional AFS storage types in the future such as NFS and Private Endpoint.
- Review and refactor Retention Agent manifests and code during this effort to improve extensibility and maintainability for future storage scenarios.

Goal #2. Vault Restores with Private Storage Accounts

- Unblock customers from using public storage accounts by enabling private storage account support for Vault restores, eliminating the need to expose List Keys for security compliance.
- Reduce customer friction around security constraints in Vault workflows.
- Multi Token Support Updates should be aligned to future extensibility for other workloads and kinds.
- This would also resolve work items for Hardening via AAD Support needed for customers data volume's security.

Goal #3. GC Notifications for Cost Awareness

- Sync with partner teams to identify and establish or reuse a mechanism to inform customers.
- Build a notification mechanism for critical cleanup workflows (GC operations) so customers are informed of failures that could impact their costs and prevent unwanted cloud expenses.
- Help establish trust with customers by providing visibility and control over maintenance operations that directly affect their spending.

Goal #4. OSS Contributions for Azure Features and Security

- Contribute design-level improvements to Velero upstream for Azure-specific features and hardening.
- Build stronger and more secure flows for Azure plugins in the upstream project.
- Maintain zero CVE posture and drive continuous dependency hygiene across Azure-managed forks.
- Work towards achieving **active contributor status**, establishing credibility and recognition in the Velero community.

Goal #5. Leveraging AI for Team Support

- Update SRE Agent Skills by Distil lessons from failures in niche scenarios, customer interactions, and error analysis into updated Wiki pages and debugging guidelines.
- Aim to reduce SME dependency for general ICMs, customer cases, and debug scenarios by making AI Skills knowledge more accessible and structured.

Goal #6. Breadth Across BCDR Workloads Beyond AKS and Disk

- Contribute to flows and features for other database workloads in BCDR (e.g., PostgreSQL, MySQL, Cosmos, Horizon DB) to break out of the AKS/Disk comfort zone.
- Gain hands-on experience in different backup and restore architectures to develop a broader systems perspective.
- Strengthen the team's cross-workload knowledge and learning across domain boundaries.

How will your actions and behaviors help you reach your goals?

- **Ownership with Learning:** While delivering AFS Vault, private SA, and GC notification workstreams, I will document decisions, trade-offs, and operating patterns so delivery and team learning happen together.
- **AI Knowledge Capture and Scaling:** I will convert lessons from incidents, customer cases, and edge failures into AI Skill for debugging guides so teams can resolve common issues without heavy SME dependency.
- **Deliberate Breadth Building:** I will contribute to BCDR workloads beyond AKS and Disk through feature work and code reviews to build stronger cross-architecture depth.

- **Proactive Collaboration:** I will sustain clear communication with partner teams, customers, and upstream maintainers through design reviews, feedback loops, and transparent status updates.
- **Upstream Contribution:** I will continue to regularly sync with maintainers, participate in discussions, and provide thoughtful PRs to build credibility upstream to strengthen Azure's technical leadership in the Velero community.

Manager comments

Looks good Priyansh.